HydroPulse-SVR: El Niño-Based KRS Reservoir Inflow Deficit Prediction Using Support Vector Regression

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***Abstract*- El Niño–Southern Oscillation (ENSO) can alter monsoon rainfall and, through it, reservoir inflow and the water available for hydropower. This paper presents HydroPulse-SVR, a pipeline that predicts the next-month inflow deficit (%) of the Krishnarajasagara (KRS) reservoir in the Cauvery basin from ENSO indices (ONI and SOI), rainfall, and reservoir inflow and storage for 2011–2020. A Support Vector Regression (SVR) model with a radial basis function (RBF) kernel is trained on a chronological split using 11 engineered features and compared with Linear Regression, Ridge Regression, Random Forest and a Persistence baseline. The predicted deficit is presented as a hydropower-risk indicator in a six-page Streamlit dashboard. On a held-out test set of 35 monthly samples, SVR-RBF obtains an RMSE of 65.5, an MAE of 53.0 and an R^2 of 0.29, with 23% of predictions within the ±20% tolerance band. It outperforms Random Forest and Persistence on RMSE and R^2, but is slightly behind Linear and Ridge regression, which suggests that the small dataset and the strong autoregressive structure limit the benefit of a non-linear kernel. The results are preliminary and are discussed together with their limitations and the planned extension to further reservoirs and longer horizons.**

***Index Terms*- El Niño, ENSO, reservoir inflow, hydropower risk, support vector regression, time-series forecasting, Streamlit.**

I. INTRODUCTION

El Niño can influence rainfall and monsoon conditions, and changes in rainfall affect river flow and reservoir inflow, which in turn determine the water available for hydropower generation. The chain El Niño → rainfall change → reservoir inflow change → hydropower risk is shown in Fig. 1. The relationship is not constant: Kumar *et al.* [1] found that the inverse relationship between ENSO and the Indian summer monsoon has weakened in recent decades, so ENSO indices alone are not a sufficient predictor of local inflow.

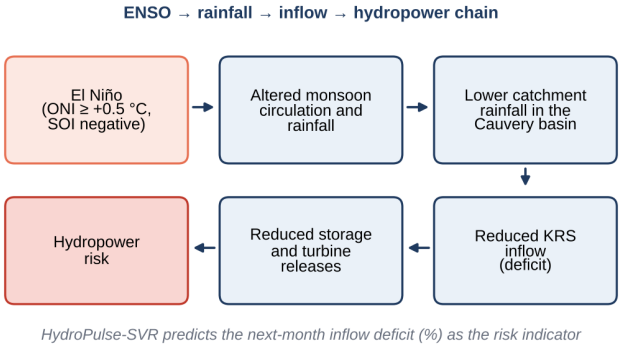


Fig. 1. From El Niño to hydropower risk (conceptual diagram).

The consequences can be large. Pulgarín-Morales and Krueger [2] report a 25–50% drop in hydropower generation in El Niño years. Reservoir inflow is hard to predict because it depends on ENSO conditions, rainfall, previous inflow, storage and seasonal variation. This work therefore asks: how can historical ENSO, rainfall and KRS reservoir data be used to predict the next-month KRS inflow deficit percentage?

The research objectives are: (RO1) to build a monthly dataset that integrates ENSO, rainfall and KRS reservoir records for 2011–2020; (RO2) to derive lag, seasonal and autoregressive features without temporal leakage; (RO3) to train an SVR-RBF model for the next-month inflow deficit and compare it with four baselines under a strict chronological protocol; and (RO4) to deliver the prediction as a hydropower-risk indicator in an interactive dashboard.

The contributions are: (i) a monthly integrated dataset combining ENSO, rainfall and KRS reservoir data; (ii) a documented feature-engineering and target-construction procedure that avoids look-ahead; (iii) an SVR-RBF model benchmarked against Linear, Ridge, Random Forest and Persistence models; and (iv) a Streamlit dashboard with six pages that converts the predicted deficit into a risk category. The remainder of the paper reviews related work (Section II), describes the data (III), methodology (IV), system (V), SVR formulation (VI), experimental setup (VII), results (VIII), dashboard (IX), limitations (X) and conclusions (XI).

II. RELATED WORK AND RESEARCH GAP

A. ENSO and the Indian Monsoon

The Indian summer monsoon varies on interannual to decadal scales, and its association with ENSO is among the most studied teleconnections [11]. Warm (El Niño) events are generally associated with weaker monsoon rainfall, but Kumar *et al.* [1] and Torrence and Webster [12] show that the ENSO–monsoon relationship has changed over time. ENSO is commonly monitored through the sea-surface-temperature-based Oceanic Niño Index (ONI) and the pressure-based Southern Oscillation Index (SOI) [13], which are the two indicators used here [7].

B. Climate Variability and Hydropower

Pulgarín-Morales and Krueger [2] quantified ENSO impacts on hydropower and proposed adaptation strategies, but did not provide a predictive model. Çakıcı [4] used LSTM and SVR to forecast hydropower generation and showed the influence of climate factors; the target was generation rather than inflow deficit and no interactive dashboard was offered.

C. Machine Learning for Reservoir Inflow and Streamflow Forecasting

Data-driven models have been used for rainfall–runoff and streamflow prediction for decades, from early neural networks [17] to the broad families of artificial-intelligence models reviewed by Yaseen *et al.* [19] and recurrent LSTM models for flood forecasting [20]. Pulgarín-Morales and Schwenzer [3] applied machine learning to daily and monthly streamflow forecasting in hydropower basins, but did not include climate indicators such as ENSO.

D. Support Vector Regression in Hydrology

SVR [6], [14], [15] is attractive for small samples because its capacity is controlled by regularisation rather than by the number of parameters. It has been applied to flood-stage forecasting [18] and, in an enhanced twin form, to time-series energy forecasting by Fu [5], without application to reservoir inflow or ENSO data.

E. Research Gap

Existing studies either quantify ENSO–hydropower impacts without prediction [2], forecast streamflow or generation without ENSO inputs or without an inflow-deficit target [3], [4], or propose models without a decision-oriented interface [5]. This work combines ENSO, rainfall and reservoir data to predict the next-month KRS inflow deficit with a standard SVR [6] and delivers the result through a dashboard with a risk indicator.

III. STUDY AREA AND DATASET

A. Study Area

The case study is the KRS reservoir in Karnataka, India, in the Cauvery basin. KRS regulates water for irrigation and municipal drinking supplies and acts as the master control reservoir whose discharge influences the downstream cascade stations Shivanasamudra (96 MW), Shimsha (17.2 MW) and Mettur Dam (200 MW).

B. Data Sources

ENSO data (ONI and SOI) come from the NOAA Climate Prediction Center [7], monthly rainfall from the World Bank Climate Change Knowledge Portal [8] (a point estimate derived from ERA5 reanalysis [25]), and KRS inflow and storage from the India Water Resources Information System (India-WRIS) [9]. The sources, resolutions and roles are summarised in Table I. The period is 2011–2020 (120 months), of which 116 samples are usable for supervised learning: 57 for training, 24 for validation and 35 for testing.

TABLE I
DATA SOURCES AND VARIABLES

Source	Variables	Native step	Role
NOAA CPC [7]	ONI, SOI	Monthly	ENSO predictors
World Bank CCKP [8]	Rainfall (mm)	Monthly	Rainfall predictor
India-WRIS [9]	Inflow, outflow, storage, level	Daily	Target and reservoir state

C. Exploratory Characteristics

Fig. 2 shows the ENSO indices over the study period. The record contains a strong El Niño in 2015–2016 (ONI peaking near +2.6) and La Niña conditions in 2011–2012 and 2017–2018, so both phases occur in the training and later periods. Of the 116 usable months, 32 are El Niño, 30 La Niña and 54 neutral.

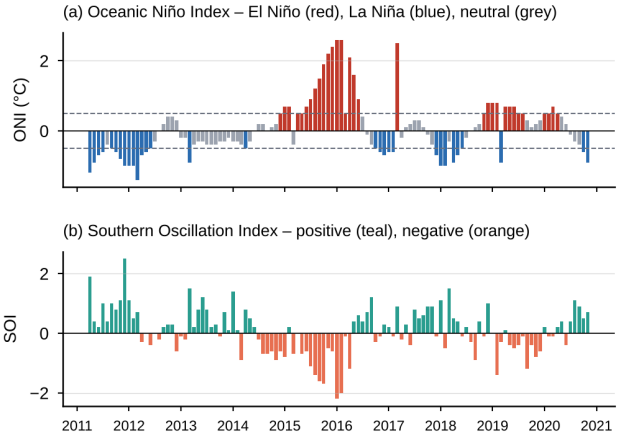


Fig. 2. Variation of (a) ONI and (b) SOI over the study period (usable months, Apr 2011–Nov 2020).

Fig. 3 shows the hydrological series. Rainfall and inflow are strongly seasonal, with inflow concentrated in the monsoon months and dominated by a few large events (a peak near 2,900 MCM in 2018), while storage cycles between roughly 13% and 100% of the observed maximum. The heavy-tailed inflow distribution motivates the use of a deficit relative to a monthly normal, rather than raw inflow, as the prediction target.

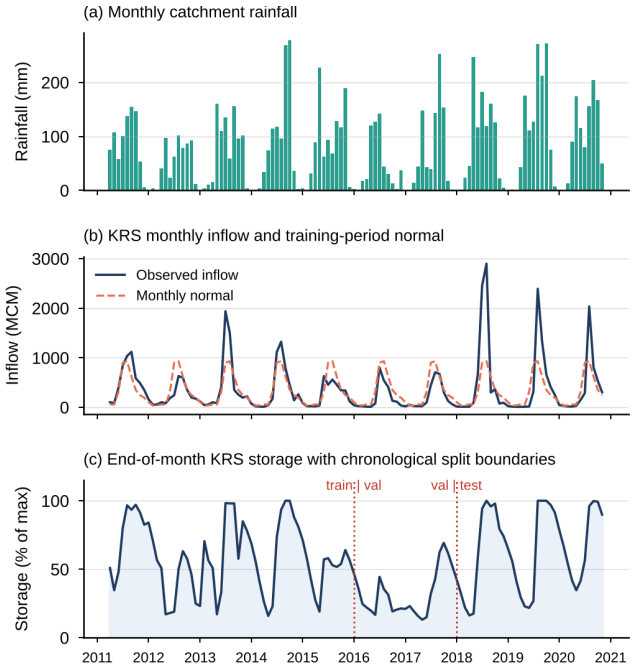


Fig. 3. (a) Monthly rainfall, (b) KRS inflow with the training-period monthly normal, and (c) end-of-month storage with the train/validation/test boundaries.

IV. METHODOLOGY: PREPROCESSING AND FEATURE ENGINEERING

A. Data Collection and Integration

The data workflow is shown in Fig. 4. Daily India-WRIS records are aggregated to monthly values: inflow and outflow in cusecs are summed over each month and converted to million cubic metres (MCM),

$$V_m = \frac{86400 \times 0.0283168 \sum_{d \in m} Q_d}{10^6} \quad (1)$$

where Q_d is the daily discharge in cusecs, 86400 is the number of seconds per day and 0.0283168 converts ft³ to m³. End-of-month storage is retained and expressed as a percentage of the maximum observed storage, which serves as a proxy for full reservoir level. ENSO, reservoir and rainfall tables are joined on a monthly period key; World Bank rainfall is preferred and the India-WRIS catchment rainfall is used as a fall-back where it is missing.

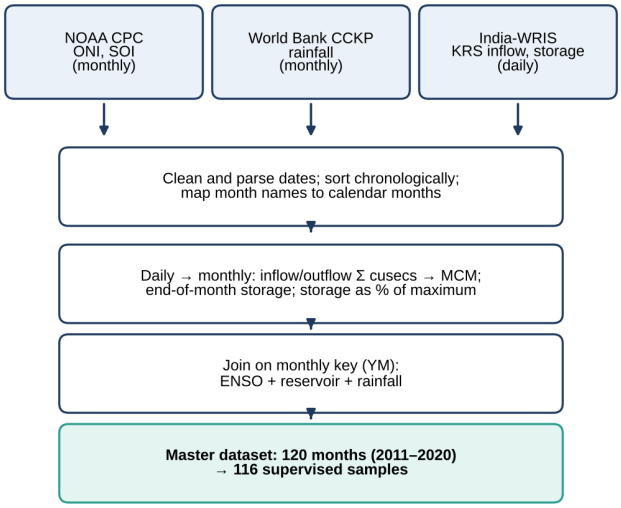


Fig. 4. Data collection and preprocessing workflow.

B. Target Variable: Next-Month Inflow Deficit

For calendar month m, the normal inflow N_m is the mean monthly inflow over the training years Y = 2011–2016, so that no test-period information enters the normals:

$$N_m = \frac{1}{|Y|} \sum_{y \in Y} V_{y,m} \quad (2)$$

The current-month inflow deficit and the prediction target are

$$D_t = \frac{N_m - V_t}{N_m} \times 100, \quad y_t = D_{t+1} \quad (3)$$

A positive deficit indicates below-normal inflow (water and power deficit risk) and a negative value indicates surplus. The target is clipped at its 2nd and 98th percentiles to limit the influence of extreme flood months.

C. Feature Engineering

Eleven features are used (Table II), a number chosen to remain safe for roughly 57 training samples. ONI lags of two and three months allow for the delay between equatorial Pacific anomalies and monsoon-season catchment response; an exploratory analysis in the project repository reports a peak correlation of about +0.29 between ONI at a three-month lag and the inflow deficit. Previous-month inflow and the current deficit supply the autoregressive signal. The calendar month is encoded cyclically so that December and January are adjacent:

$$s_t = sin(2\pi m_t/12), \quad c_t = cos(2\pi m_t/12) \quad (4)$$

All shifts use only past values, so no feature contains information from month t+1. The first three rows (lost to lags) and the last row (no t+1 target) are dropped, which reduces 120 monthly observations to 116 samples. The prediction pipeline is illustrated in Fig. 5.

TABLE II
FEATURE SET (11 FEATURES)

Feature	Description
ONI, SOI	Oceanic Niño Index; Southern Oscillation Index at month t
ONI_{lag2}, ONI_{lag3}	ONI two and three months earlier
rainfall_mm	Monthly catchment rainfall (mm)
inflow_mcm	Current-month KRS inflow (MCM)
inflow_lag1_mcm	Previous-month inflow (MCM)
storage_pct_frl	End-of-month storage, % of maximum
month_sin, month_cos	Cyclic encoding of calendar month
deficit_pct	Current-month inflow deficit D_t (%)

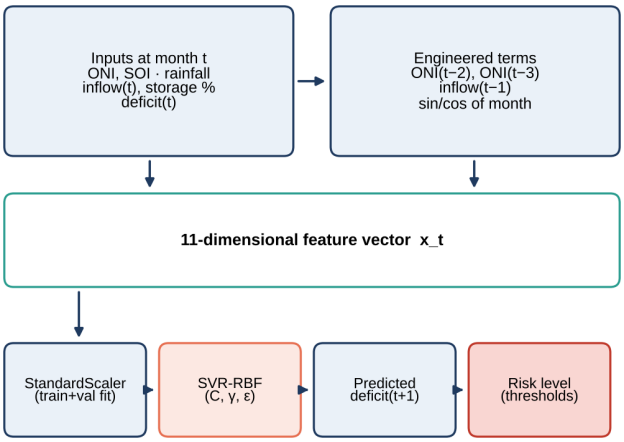


Fig. 5. Feature engineering and prediction pipeline.

D. Chronological Split and Standardisation

Data are split strictly in time order (Table III) to mimic operational forecasting. Features are standardised with StandardScaler,

$$z = (x - \mu)/\sigma \tag{5}$$

with μ and σ estimated from the training and validation samples only; test samples are transformed but never used to fit the scaler.

TABLE III
CHRONOLOGICAL TRAIN–VALIDATION–TEST SPLIT

Subset	Period	Samples
Training	Apr 2011 – Dec 2015	57
Validation	Jan 2016 – Dec 2017	24
Testing	Jan 2018 – Nov 2020	35
Total usable		116

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

HydroPulse-SVR is organised in four layers (Fig. 6). The data layer holds the ENSO, rainfall and reservoir sources. The preprocessing layer is implemented in *config.py*, which centralises paths, split boundaries, the feature list and the target name, and in *data_pipeline.py*, which loads, aggregates, merges and engineers features and writes the master dataset. The model layer, *train_model.py*, performs the chronological split, scaling, SVR tuning, baseline training and evaluation, and persists the artefacts (*svr_model.pkl*, *scaler.pkl*, *master_data.csv*, *metrics.csv*, *predictions.csv*). The presentation layer, *dashboard.py*, is a Streamlit application [24] that reads these artefacts and exposes six pages with interactive Plotly charts.

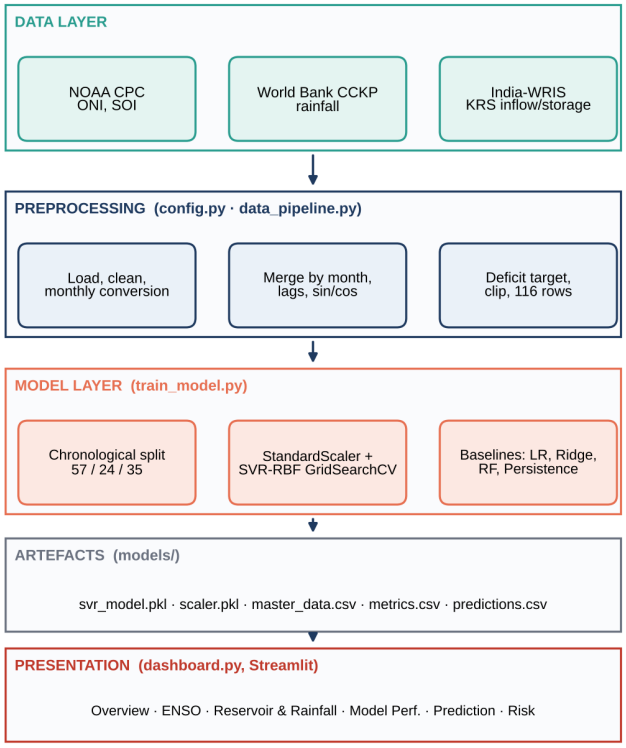


Fig. 6. Complete HydroPulse-SVR system architecture.

The implementation uses Python with pandas, NumPy, scikit-learn [10] and joblib. Training is deterministic apart from the Random Forest baseline, whose random seed is fixed. The end-to-end procedure is summarised in Algorithm 1.

Algorithm 1: HydroPulse-SVR training and prediction
1: Load ONI/SOI, daily KRS data and monthly rainfall; convert to monthly (1).
2: Merge on month; compute normals N_m from 2011–2016 (2).
3: Compute D_t, target $y_t = D_{t+1}$ (3); build 11 features (4); clip target; drop incomplete rows.
4: Split chronologically into train / validation / test (Table III).
5: Fit StandardScaler on train + validation (5).
6: Tune (C, γ, ε) by GridSearchCV with TimeSeriesSplit on the training set (10).
7: Refit SVR and baselines on train + validation.
8: Predict the test set; compute RMSE, MAE, R^2, ±20% share.
9: Persist model, scaler, metrics, predictions; serve via Streamlit; map prediction to a risk level (Table VII).

VI. SUPPORT VECTOR REGRESSION FORMULATION

Given training pairs $(\mathbf{x}_i, y_i)$, $i = 1,\ldots,n$, ε-SVR seeks a function $f(\mathbf{x})$ that deviates from the targets by at most ε

wherever possible while remaining as flat as possible [6]. The ε-insensitive loss ignores errors inside a tube of half-width ε:

$$|\xi|_{\varepsilon} = max(0, |\xi| - \varepsilon) \quad (6)$$

The primal problem, with slack variables ξ_i, ξ_i^* for points outside the tube, is

$$min \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \quad (7)$$

subject to $y_i - f(\mathbf{x}_i) \le \varepsilon + \xi_i$, $f(\mathbf{x}_i) - y_i \le \varepsilon + \xi_i^*$, and $\xi_i, \xi_i^* \ge 0$. The regularisation constant C trades flatness against tolerated deviations. In the dual form the regressor is

$$f(x) = \sum_{i=1}^{n}(\alpha_i - \alpha_i^*)K(x_i, x) + b \quad (8)$$

where only support vectors have non-zero coefficients $\alpha_i - \alpha_i^*$. With the RBF kernel,

$$K(x, x') = exp(-\gamma\|x - x'\|^2) \quad (9)$$

non-linear relationships between the standardised features and the target can be modelled; larger γ gives a more local, flexible fit, while larger ε gives a sparser model with a wider insensitive tube. Standardisation (5) is essential because the kernel depends on Euclidean distances. Hyper-parameters (C, γ, ε) are tuned with GridSearchCV over TimeSeriesSplit folds, implemented in scikit-learn [10],

$$\theta^* = arg\,min_{\theta}\frac{1}{K}\sum_{k=1}^{K}RMSE_k(\theta) \quad (10)$$

with $K = 4$ expanding-window folds. The SVR objective differs from the squared-error criterion of RMSE, because errors inside the ε-tube are ignored and large errors are penalised linearly rather than quadratically; this matters when the models are compared on RMSE and R^2.

VII. EXPERIMENTAL SETUP AND EVALUATION METRICS

The training and evaluation workflow is shown in Fig. 7, and the configuration is given in Table IV. SVR hyper-parameters are tuned only on the training portion; the final SVR and the baselines are refitted on training plus validation data and evaluated once on the untouched test period 2018–2020.

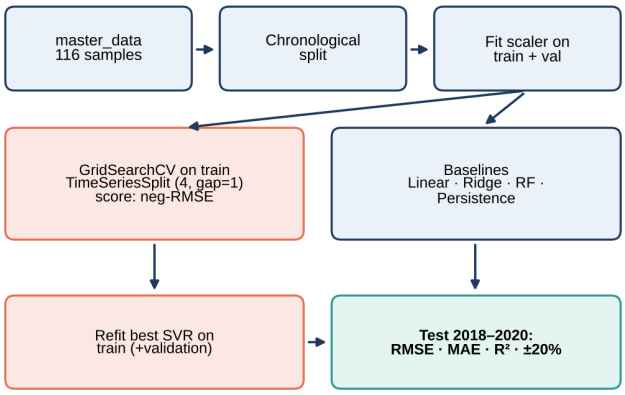


Fig. 7. Model training and evaluation workflow.

TABLE IV
EXPERIMENTAL CONFIGURATION

Item	Setting
SVR-RBF	C ∈ {10, 50, 100, 200}; ε ∈ {0.5, 1, 2, 5}; γ ∈ {0.005, 0.01, 0.02, 0.05, 0.1, scale}
Tuning	GridSearchCV, TimeSeriesSplit (4 folds, gap = 1 month), score: negative RMSE
Linear	Ordinary least squares on the 11 standardised features
Ridge	Regularisation strength 1.0, Eq. (11)
Random Forest	100 trees, fixed random seed (42) [22]
Persistence	$\hat{y}_{t+1} = D_t$ (current deficit carried forward)

Ridge Regression [21] minimises a penalised squared error,

$$min \|y - Xw\|^2 + \lambda\|w\|^2 \quad (11)$$

Performance is measured on the test set by the root-mean-square error, mean absolute error and coefficient of determination,

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (12)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (13)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (14)$$

The dashboard additionally reports the share of test predictions whose absolute error does not exceed 20 percentage points of deficit,

$$P_{20} = \frac{1}{n}\sum_{i=1}^{n}1[|y_i - \hat{y}_i| \le 20] \quad (15)$$

RMSE penalises large errors more than MAE, so a gap between the two indicates occasional large errors. As the temporal ordering of the data is respected, out-of-sample evaluation follows accepted practice for forecasting [23].

VIII. RESULTS AND DISCUSSION

A. Test-Set Performance

Table V lists test-set performance on the 35 held-out samples. SVR-RBF achieves RMSE 65.5, MAE 53.0 and R^2 0.29, and 23% of its test predictions fall within the ±20% tolerance band reported by the dashboard. Fig. 8 compares the models.

TABLE V
TEST-SET PERFORMANCE OF THE MODELS

Model	RMSE	MAE	R^2
SVR-RBF	**65.5**	**53.0**	**0.29**
Linear Regression	60.5	49.3	0.40
Ridge Regression	60.6	49.5	0.39
Random Forest	69.1	57.6	0.21
Persistence	77.5	48.7	0.01

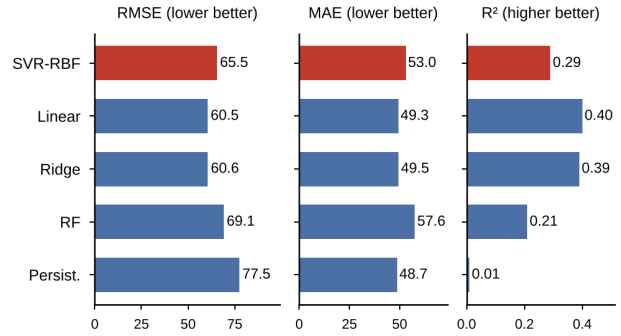


Fig. 8. Model comparison on RMSE, MAE and R^2 (RF: Random Forest; SVR-RBF in red). Values from Table V.

B. Discussion

SVR-RBF performs better than Random Forest and Persistence on RMSE and R^2. Persistence has the lowest MAE (48.7) but the highest RMSE, which suggests that it is accurate in many months but makes occasional large errors when inflow changes abruptly. Linear and Ridge regression are slightly better than SVR on all three metrics, and are almost indistinguishable from each other (RMSE 60.5 and 60.6), which indicates that the L2 penalty has little effect at this sample size. All learning models reach a positive R^2, indicating that ENSO and hydrological features carry useful information about the next-month deficit; at the same time, the best R^2 of 0.40 means that most of the variance remains unexplained.

Several factors may explain why the linear models edge out the non-linear kernel. These are plausible explanations rather than tested conclusions. (1) *Sample size:* with 57 training samples, 11 features and three tuned hyper-parameters, an RBF model has more freedom to fit noise, and the folds used for time-series tuning are very small. (2) *Dominant linear signal:* the current deficit and previous inflow are strongly and roughly linearly related to the next-month deficit, so much of the predictable structure is captured by a linear map. (3) *Loss mismatch:* the ε-insensitive, linearly penalised SVR objective is not aligned with the squared-error RMSE and R^2 used for comparison. (4) *Non-stationarity:* the test period contains extreme events, including the 2018 flood-season inflow, that are poorly represented in 2011–2015 training data, and kernel models do not extrapolate far beyond the training domain. The RMSE difference between SVR and Linear Regression (5.0 units, about 8%) was not subjected to a significance test and may lie within sampling variability for 35 test samples. Random Forest, similarly, cannot extrapolate beyond the target values seen in training, which is a possible reason for its weaker result.

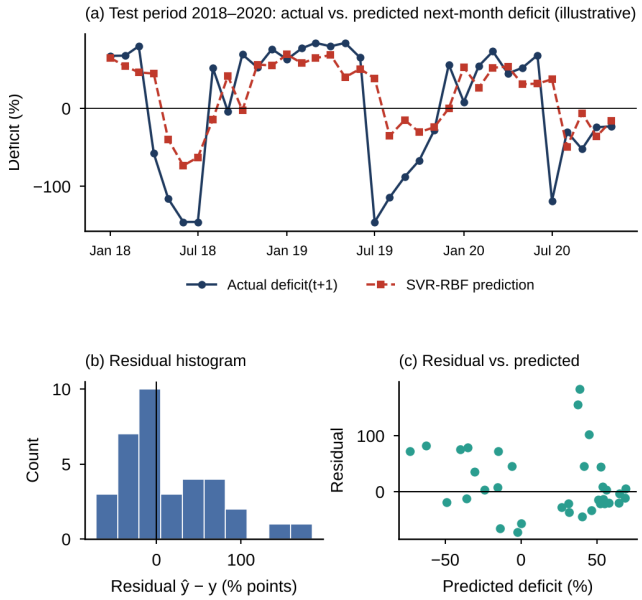


Fig. 9. Illustrative (a) actual vs. predicted next-month deficit on the test period, (b) residual histogram, (c) residuals vs. predictions. Plotted from the repository's saved prediction file; summary metrics are those of Table V.

Fig. 9 illustrates the behaviour of the SVR predictions. The predicted series follows the general direction of the actual deficit but under-represents the deepest surplus months (large negative deficits), and the residuals are spread widely with a heavier tail on the over-prediction side. This pattern is consistent with the gap between RMSE and MAE in Table V and with the modest R^2. The 23% within-tolerance figure shows that point predictions should be read as a coarse risk indicator rather than as precise deficit values, which motivates the categorical presentation in Section IX.

IX. STREAMLIT DASHBOARD AND HYDROPOWER RISK ANALYSIS

The dashboard (Fig. 10 and Fig. 11, Table VI) turns the trained model into a decision-support interface. The *Overview* page summarises ENSO state, reservoir condition and model results. *ENSO Analysis* plots ONI and SOI. *Reservoir & Rainfall* presents inflow, storage and rainfall series and monthly climatology. *Model Performance* compares SVR-RBF with the baselines on the test period (2018–2020). *Deficit Prediction* is a what-if tool in which the user adjusts ONI, SOI, rainfall, inflow, storage and month, and the stored scaler and SVR model return the next-month deficit. *Risk Analysis* categorises historical deficits, shows seasonal patterns and examines El Niño episodes.

TABLE VI
DASHBOARD PAGES

Page	Main content
Overview	KPIs, deficit series, ENSO-phase counts
ENSO Analysis	ONI by phase, SOI, ONI-lag relationship
Reservoir & Rainfall	Inflow, storage, rainfall climatology
Model Performance	Metrics table, model comparison, predictions
Deficit Prediction	What-if inputs, predicted deficit, risk badge
Risk Analysis	Risk-class counts, seasonality, El Niño impacts

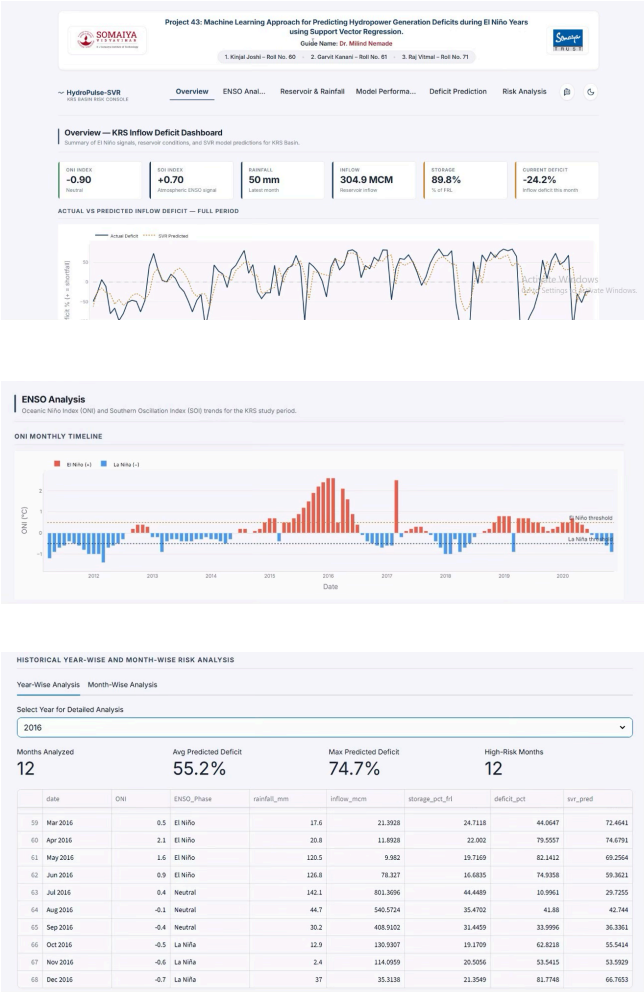


Fig. 10. Illustrative mockups (not actual screenshots) of the (a) Overview, (b) ENSO Analysis and (c) Reservoir & Rainfall pages, based on the implemented dashboard structure.

The predicted deficit is mapped to a risk category using the thresholds in Table VII. For instance, a predicted deficit of +45% is displayed as High Risk. These thresholds are those implemented in the dashboard; they are heuristic cut-offs on the deficit rather than calibrated against generation losses.

TABLE VII
DEFICIT-TO-RISK MAPPING IN THE DASHBOARD

Predicted deficit (%)	Category	Level
< 0	Surplus	0
0 to < 20	Low Risk	1
20 to < 40	Moderate Risk	2
40 to < 60	High Risk	3
≥ 60	Severe Risk	4



Fig. 11. Illustrative mockups (not actual screenshots) of the (a) Model Performance, (b) Deficit Prediction and (c) Risk Analysis pages. Input values in (b) are illustrative.

In terms of hydropower, the deficit is an indicator of upstream water scarcity at the master control reservoir. Because KRS discharge feeds the Shivanasamudra (96 MW), Shimsha (17.2 MW) and Mettur (200 MW) stations downstream, a predicted high deficit signals a higher likelihood of reduced water availability for these stations. The dashboard does not forecast generation, since actual hydropower-generation data have not yet been incorporated.

X. LIMITATIONS

The results should be read in light of the following limitations. (1) *Small dataset:* only 116 monthly samples (57 for training) are available, which limits non-linear models and the reliability of the hyper-parameter search. (2) *Limited forecasting period:* only the next month is predicted and only one test period (2018–2020) is evaluated; results over multiple test periods are not yet available. (3) *Complex ENSO–rainfall relationship:* the ENSO–monsoon link is non-stationary [1], [12], and no feature-importance or ablation analysis has yet isolated the contribution of the ENSO features. (4) *Additional reservoir data:* only KRS is studied, and further storage, release and basin-wide data are needed. (5) *Hydropower-generation data:* risk categories are based on deficit thresholds and are not validated against actual generation. (6) *Model refinement:* the final selected hyper-parameter values, hyper-parameter sensitivity, and a statistical comparison of models remain to be reported. (7) *Pipeline details:* monthly normals are computed over 2011–2016, which extends one year into the validation period, and the target-clipping percentiles are computed on the whole sample; both are minor but should be tightened in future versions. (8) *Illustrative outputs:* the dashboard figures are mockups, and Fig. 9 is plotted from a saved prediction file.

XI. CONCLUSION AND FUTURE SCOPE

A complete pipeline for KRS next-month inflow-deficit prediction has been built, combining ENSO, rainfall and reservoir data with an SVR-RBF model and a six-page Streamlit dashboard. On 35 test samples SVR-RBF reaches RMSE 65.5, MAE 53.0 and R^2 0.29, ahead of Random Forest and Persistence on RMSE and R^2 but slightly behind Linear and Ridge regression, so the relationship appears largely linear at this data size. Results are preliminary and limited by the small dataset.

Future work includes extending the pipeline to the Harangi, Kabini and Hemavathi reservoirs, assembling a larger historical dataset, adding actual hydropower-generation data, evaluating longer forecasting horizons, and testing LSTM [16], [20] and other advanced models, together with feature-importance analysis, hyper-parameter sensitivity and multi-period evaluation.

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